

Solving algebraic expression with exponential functions

$y = 5^x$ is an example of an exponential function.

$25^x = 125$ is an example of an exponential equation.

When solving exponential equations, it is important to identify the base number (usually a prime number) that is common to all the individual terms of the equation.

For example,

$$25^x = 125; 5 \text{ is the base number}$$

$$7^x = 2401; 7 \text{ is the base number}$$

Worked example:

$$\frac{1}{8^x} = 32^{\frac{1}{3}}$$

$$\frac{1}{(2^3)^x} = (2^5)^{\frac{1}{3}}$$

$$2^{-3x} = 2^{\frac{5}{3}}$$

$$-3x = \frac{5}{3}$$

$$-x = \frac{5}{9}$$

$$x = -\frac{5}{9}$$

The base number is 2

Worked example 2:

If $27^a = 9$ and $2^{a-b} = 64$, find the value for a and b

So taking one term at a time we will find value for a

$$27^a = 9$$

$$3^{3a} = 3^2$$

$$3a = 2$$

$$a = \frac{2}{3}$$

The base number is 3

Knowing that $a = \frac{2}{3}$ we can substitute it into the other equation above

$$2^{a-b} = 64$$

$$2^{\frac{2}{3}-b} = 64$$

$$2^{\frac{2}{3}-b} = 2^6$$

$$\frac{2}{3} - b = 6$$

$$-b = 6 - \frac{2}{3}$$

$$-b = \frac{16}{3}$$

$$b = -\frac{16}{3}$$

The base number is 2

So the value for $a = \frac{2}{3}$ and $b = -\frac{16}{3}$

Worked example 3:

Solve the following equation for x:

$$\frac{3}{7}x^2 - 1 = 5$$

$$\frac{3}{7}x^2 = 5 + 1$$

$$\frac{3}{7}x^2 = 6$$

$$3x^2 = 6 \times 7$$

$$3x^2 = 42$$

$$x^2 = \frac{42}{3}$$

$$x^2 = 14$$

$$x = \sqrt{14}$$

$$x = \pm 3.74$$